

1. KNN

2. Gradient Descent / Gradient Descent ($f(w) = -f(w)$)
change the function to a convex.

3. polynomial kernel

4. $P(B|A, C) = P(B|C)$ as A, B are independent.

5. _____

6. L1 Regression or L2 Regression

7. K-means

8. Sigmoid

9. MSE

10.

11 a. $FPR = \frac{FP}{FP+TN}$

b. $F_2 = \frac{(2^2+1)P \cdot R}{2^2 \cdot P + R} = \frac{5P \cdot R}{4P + R}$

12 a. Underfitting

b. Overfitting

13.	a	b	c	d	e	f	g	h	i	j
	F	F	T	T	F	F	T	T	F	T

1. initial $w = [0, 0]$

$$\theta_0 = [0, 0]$$

$$y_1 = 0 \times 2 + 0 \times 1 = 0 \geq 0 \quad + \quad \text{True}$$

$$y_2 = 0 \times (-2) + 0 \times 3 = 0 \geq 0 \quad + \quad \text{True}$$

$$y_3 = 0 \times (-4) + 0 \times 0 = 0 \geq 0 \quad - \quad \text{False refresh } \theta_0$$

$$\theta_1 = \theta_0 - [-4, 0] = [4, 0]$$

$$y_4 = 4 \times 1 + 0 \times (-2) = 4 \geq 0 \quad - \quad \text{False refresh } \theta_1$$

$$\theta_2 = \theta_1 - [1, -2] = [3, 2]$$

$$y_5 = 3 \times 1 + 2 \times 2 = 7 \geq 0 \quad + \quad \text{True}$$

$$y_6 = 3 \times (-3) + 2 \times (-2) = -13 < 0 \quad - \quad \text{True}$$

$$\hookrightarrow y_1 = 3 \times 2 + 2 \times 1 = 8 \geq 0 \quad + \quad \text{True}$$

$$y_2 = 3 \times (-2) + 2 \times 3 = 0 \geq 0 \quad + \quad \text{True}$$

$$y_3 = 3 \times (-4) + 2 \times 0 = -12 < 0 \quad - \quad \text{True}$$

$$y_4 = 3 \times 1 + 2 \times (-2) = -1 < 0 \quad - \quad \text{True}$$

$$w = [3, 2]$$

$$\text{separator: } 3x_1 + 2x_2 = 0$$

2

3	2	3	2	
Age	Sex	BP	Cholesterol	
				A
				B

$$P(A) = \frac{5}{14}$$

$$P(B) = \frac{9}{14}$$

$$\frac{7}{14}$$

$$P(\text{Young} | A) = \frac{3+1}{5+3} = \frac{4}{8} = \frac{1}{2}$$

$$P(\text{Young} | B) = \frac{2+1}{9+3} = \frac{3}{12} = \frac{1}{4}$$

$$P(\text{Male} | A) = \frac{1+1}{5+3} = \frac{2}{8}$$

$$P(\text{Male} | B) = \frac{6+1}{9+3} = \frac{7}{12}$$

$$P(\text{High} | A) = \frac{2+1}{5+3} = \frac{3}{8}$$

$$P(\text{High} | B) = \frac{2+1}{9+3} = \frac{3}{12}$$

$$P(\text{Unsafe} | A) = \frac{3+1}{5+3} = \frac{4}{8}$$

$$P(\text{Unsafe} | B) = \frac{3+1}{9+3} = \frac{4}{12}$$

$$\begin{aligned}
 P(A|—) &= P(A) \cdot \cdot \cdot \cdot \\
 &= \frac{5}{14} \times \frac{1}{2} \times \frac{2}{7} \times \frac{3}{8} \times \frac{4}{7} = \boxed{\frac{120}{128}} \\
 &= 0.01093 \\
 &\approx 0.011
 \end{aligned}$$

$$\begin{aligned}
 P(B|—) &= P(B) \cdot \cdot \cdot \cdot \\
 &= \frac{9}{14} \times \frac{1}{4} \times \frac{7}{11} \times \frac{3}{12} \times \frac{4}{11} = \boxed{0.00929} \\
 &\approx 0.009
 \end{aligned}$$

arranged $P(A) - P(B) -$

$$= P(A)$$

$$0.011 / (0.011 + 0.009) = 0.55$$

~~Drug A~~.

3 - : Negative + : Positive 0 : Neutral

$$P(-) = \frac{1}{3}$$

$$P(+) = \frac{1}{3}$$

$$P(0) = \frac{1}{3}$$

27 words

15 unique words

$$P(\text{Winter} | -) = \frac{0+1}{9+15} = \frac{1}{24}$$

$$P(\text{Winter} | +) = \frac{1+1}{9+15} = \frac{2}{24}$$

$$P(\text{Winter} | 0) = \frac{1+1}{9+15} = \frac{2}{24}$$

$$P(is|-) = \frac{2+1}{9+15} = \frac{3}{24}$$

$$P(is|+) = \frac{3}{24}$$

$$P(is|0) = \frac{3}{24}$$

$$P(not|-) = \frac{2}{24}$$

$$P(not|+) = \frac{1}{24}$$

$$P(not|0) = \frac{1}{24}$$

$$P(pleasant|-) = \frac{1}{24}$$

$$P(pleasant|+) = \frac{2}{24}$$

$$P(pleasant|0) = \frac{1}{24}$$

$$P(-|0) = \frac{1}{3} \times \frac{1}{24} \times \frac{3}{24} \times \frac{2}{24} \times \frac{1}{24} = \cancel{0.000077} 6 \times 10^{-6}$$

$$P(+|0) = \frac{1}{3} \times \frac{2}{24} \times \frac{3}{24} \times \frac{1}{24} \times \frac{2}{24} = \cancel{0.000077} 12 \times 10^{-6}$$

$$P(0|0) = \frac{1}{3} \times \frac{2}{24} \times \frac{3}{24} \times \frac{1}{24} \times \frac{1}{24} = \cancel{0.000077} 6 \times 10^{-6}$$

~~The sentence is Negative or Neutral.~~

The sentence is positive

4. $\eta = 1.0$

$\theta_0 = [-0.3, 0.7, 0.5]$

	$\theta^T x$	$\theta^T x - y_i$	$(\theta^T x - y_i) x_i$
x_1	$0.3 - 2.1 + 1$ -0.8	$-0.8 + 1.3$ 0.5	
x_2	$1.4 - 0.5$ 0.9	$0.9 - 1.7$ -0.8	
x_3	$-0.9 - 0.7 + 0.5$ -1.1	$-1.1 - 2.5$ -3.6	

$\frac{3}{16} \times \frac{1}{3}$

$\theta_0 = \frac{\eta}{n} \sum_{i=1}^3 x_{ij} (\theta^T x - y_i)$

$\frac{2.4}{1} \times \frac{1}{3}$

$w_1 = \frac{1}{3} [-1 \times 0.5 + 0 \times (-0.8) + 3 \times (-3.6)] = -3.77$

-2

$w_2 = \frac{1}{3} [-3 \times 0.5 + 2 \times (-0.8) + -1 \times (-3.6)] = 0.17$

$w_3 = \frac{1}{3} [-1.3 \times 0.5 + 1.7 \times (-0.8) + 2.5 \times (-3.6)] = -11.01$

$$\begin{bmatrix} -0.3 \\ 0.7 \\ 0.5 \end{bmatrix} - \begin{bmatrix} -3.77 \\ 0.17 \\ -3.67 \end{bmatrix} = \begin{bmatrix} 3.47 \\ 0.53 \\ 4.17 \end{bmatrix}$$

-3.67

RMS ~~in~~ not calculated. No time left.